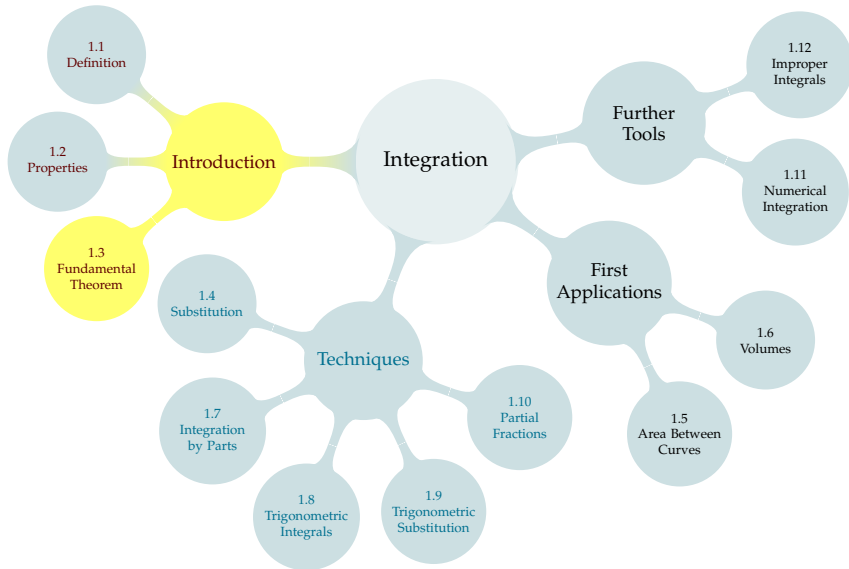


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- ▶ Limit of a Riemann Sum

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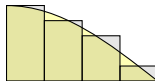
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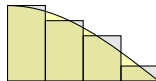
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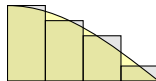
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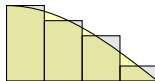
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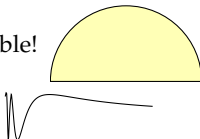
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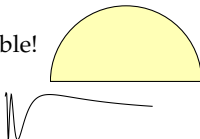
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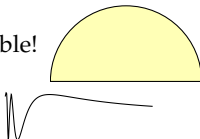
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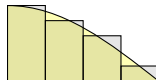
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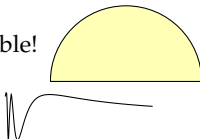
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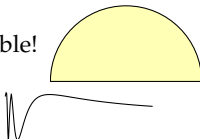
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- ▶ Doesn't work for every function

Fundamental Theorem of Calculus, Part 1

Let $a < b$ and let $f(x)$ be a function which is defined and continuous on $[a, b]$. Let

$$A(x) = \int_a^x f(t) \, dt$$

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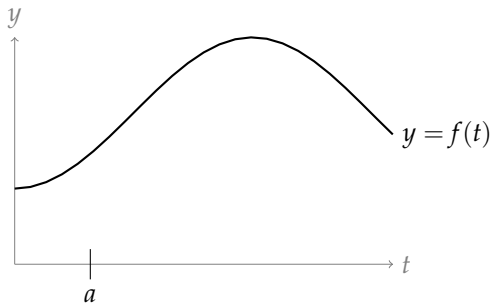
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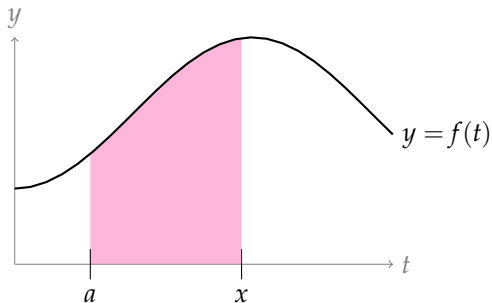
FTC(I) gives us the derivative of a very specific function (subject to some fine print).

It shows a close relationship between integrals and derivatives.

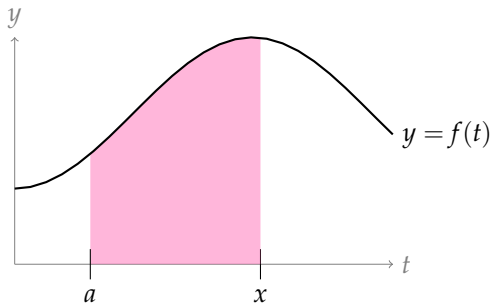
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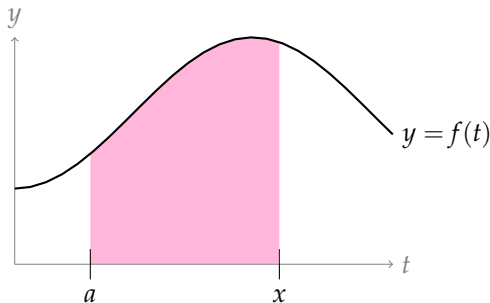
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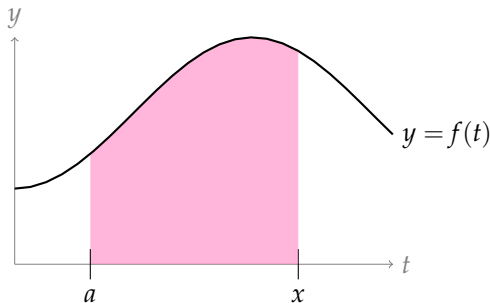
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Notation: the function A depends on the variable x .

We need to know how the function f behaves on the whole interval $(0, x)$ to find $A(x)$. That's why we use $f(t)$, not $f(x)$.

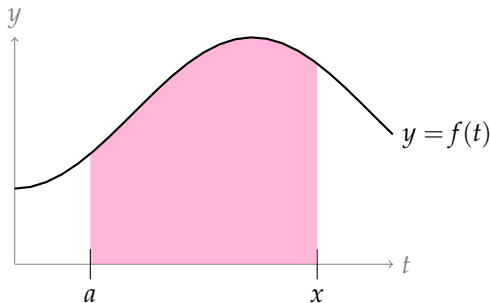
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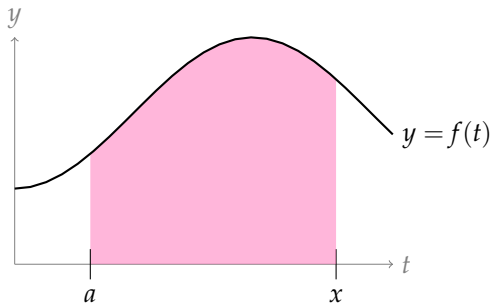
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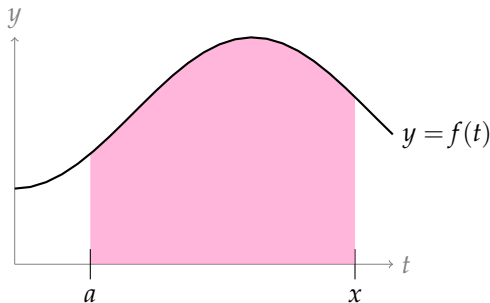
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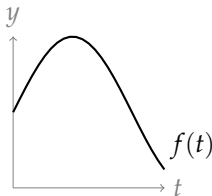
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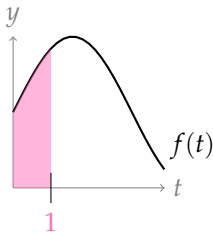


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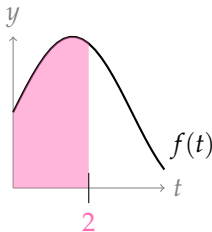
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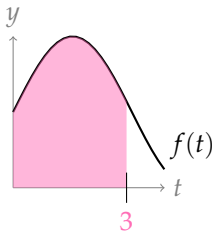
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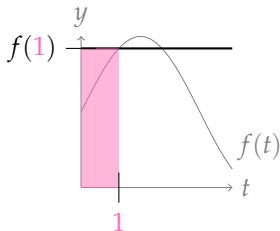
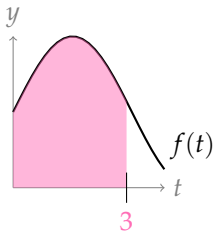
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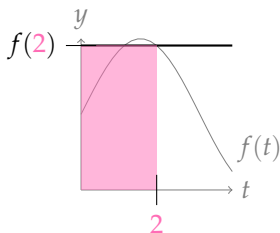
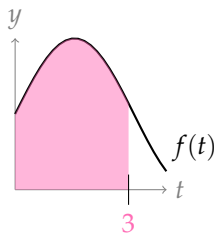
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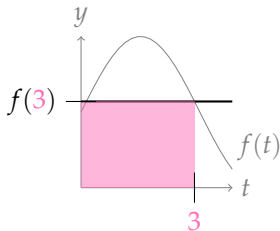
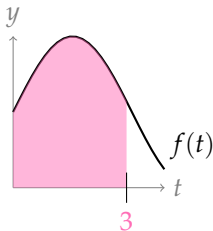
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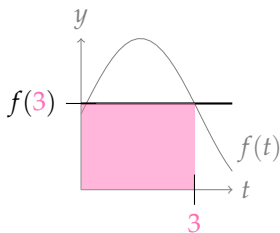
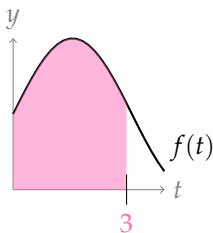
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$$C(1) = \int_0^1 f(1) \underbrace{dx}_{??}$$



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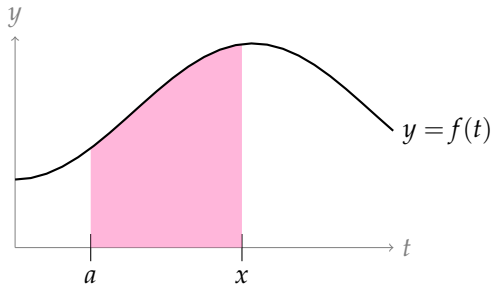
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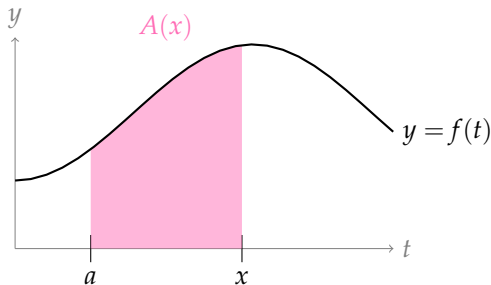
Question: Why is it true?

DERIVATIVE OF AREA FUNCTION, $A(x) = \int_a^x f(t) dt$



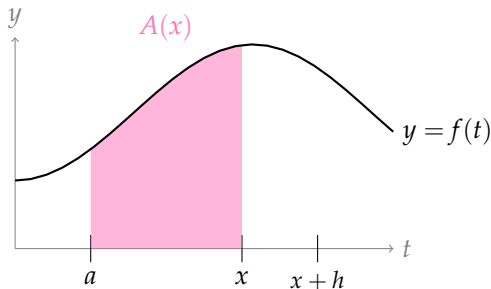
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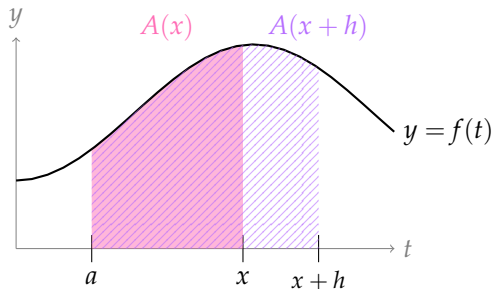
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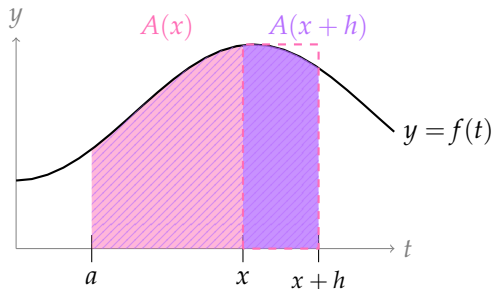
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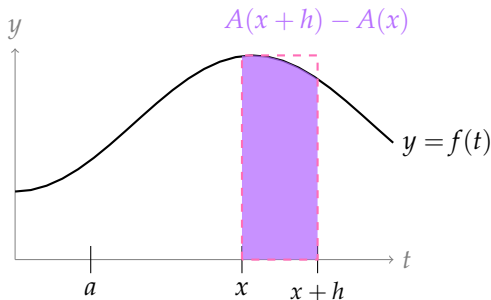
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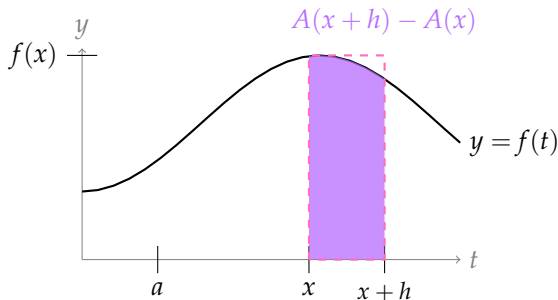
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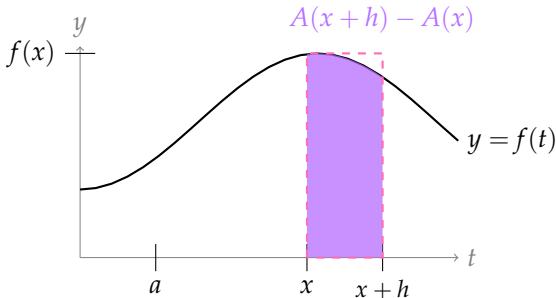
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When h is very small, the purple area looks like a rectangle with base h and height $f(x)$, so $A(x+h) - A(x) \approx hf(x)$ and $\frac{A(x+h) - A(x)}{h} \approx f(x)$. As h tends to zero, the error in this approximation approaches 0.

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$$B'(x) = \frac{d}{dx} \left\{ - \int_2^x f(t) dt \right\} = - \sin x$$

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Suppose $C(x) = \int_2^{e^x} \sin t \, dt$. What is $C'(x)$?



Fundamental Theorem of Calculus, Part 1

Let $a < b$ and let $f(x)$ be a function which is defined and continuous on $[a, b]$. Let

$$A(x) = \int_a^x f(t) \, dt$$

for any x in $[a, b]$. Then the function $A(x)$ is differentiable and

$$A'(x) = f(x) .$$

Suppose $C(x) = \int_2^{e^x} \sin t \, dt$. What is $C'(x)$?

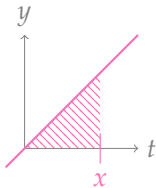
$C'(x) = e^x \sin(e^x)$: if we set $a = 2$, then

$$\begin{aligned} C(x) &= A(e^x) \\ \implies C'(x) &= A'(e^x) \cdot \frac{d}{dx}\{e^x\} = \sin(e^x) \cdot e^x \end{aligned}$$

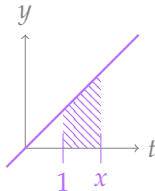


It's possible to have two different functions with the same derivative.

$$A(x) = \int_0^x 2t \, dt$$

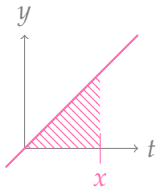


$$B(x) = \int_1^x 2t \, dt$$



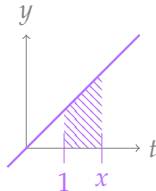
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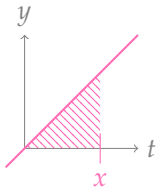
$$A'(x) = 2x$$

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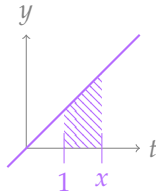
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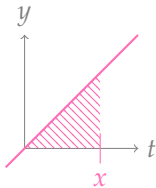
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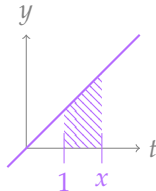
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$$A(x) = \int_0^x 2t \, dt = x^2$$



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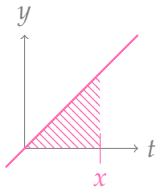
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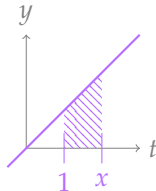
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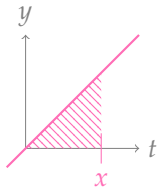
$$B(x) = \int_1^x 2t \, dt = x^2 - 1$$



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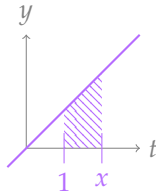
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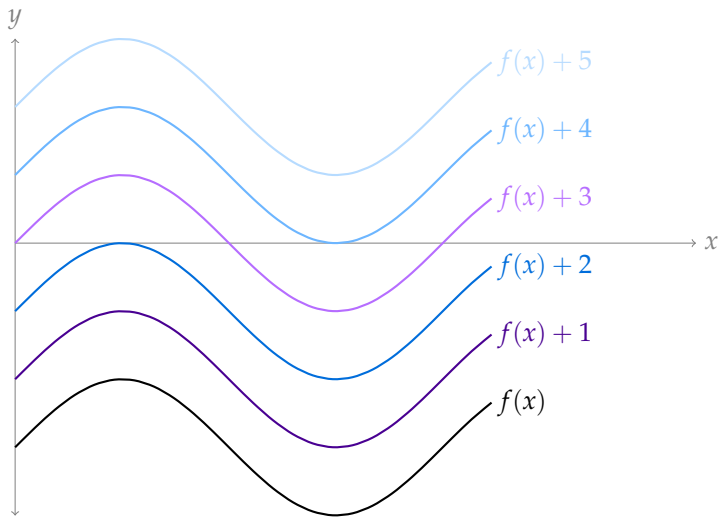
$$B(x) = \int_1^x 2t \, dt = x^2 - 1$$



$$B'(x) = 2x$$

When two functions have the same derivative, they differ only by a constant.

In this example: $B(x) = A(x) - 1$



If two continuous functions have the same derivative, then one is a constant plus the other.

Two clues for finding $A(x) = \int_a^x f(t) \, dt$:

- ▶ If $A(x) = \int_a^x f(t) \, dt$, then¹ $A'(x) = f(x)$
- ▶ If $F'(x) = A'(x)$, then $A(x) = F(x) + C$ for some constant C .

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- ▶ $A'(x) = \cos x$.
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- ▶ Then $A(x) = \sin x + C$ for some constant C .

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$A(x) = \int_{-2}^x 5t^4 \, dt$. What functions could $A(x)$ be?

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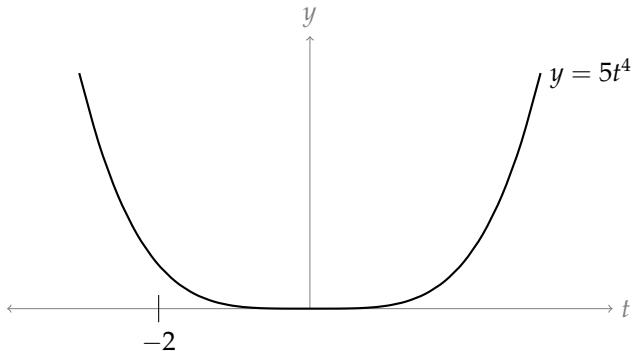
- ▶ $A'(x) = 5x^4$.
- ▶ Guess a function with derivative $5x^4$: $F(x) = x^5$.
- ▶ Then $A(x) = x^5 + C$ for some constant C .
- ▶ We ALSO know $A(-2) = \int_{-2}^{-2} 5t^4 dt = 0$, so we can find C :

$$0 = A(-2) = (-2)^5 + C \implies C = 32$$

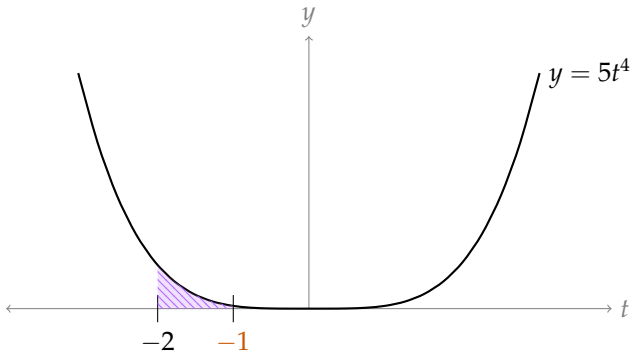
- ▶ So, $A(x) = x^5 + 32$

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$$A(x) = \int_{-2}^x 5t^4 dt = x^5 + 32$$

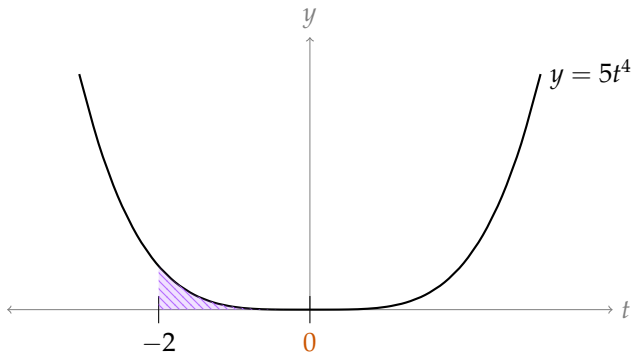


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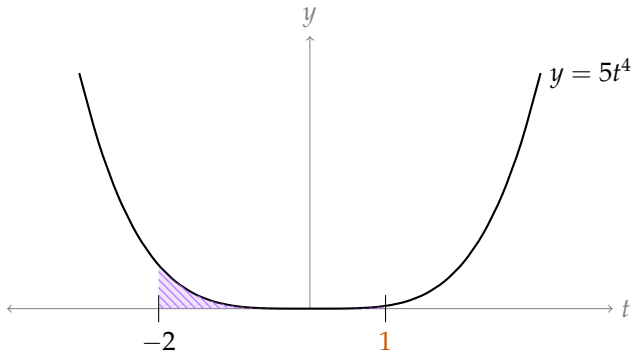
$$A(-1) = \int_{-2}^{-1} 5t^4 dt = (-1)^5 + 32 = 31$$

$$A(x) = \int_{-2}^x 5t^4 dt = x^5 + 32$$



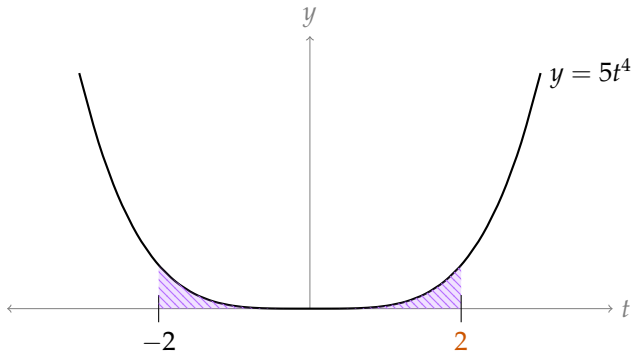
$$A(0) = \int_{-2}^0 5t^4 dt = (0)^5 + 32 = 32$$

$$A(x) = \int_{-2}^x 5t^4 dt = x^5 + 32$$



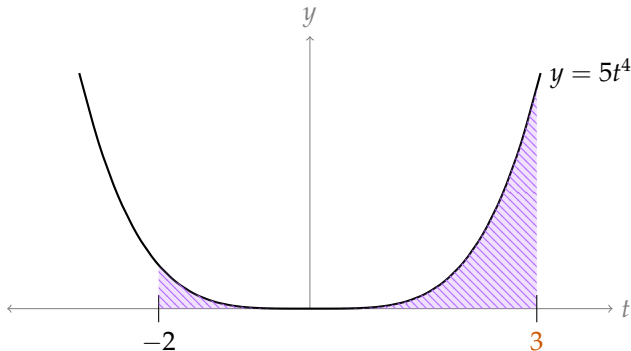
$$A(1) = \int_{-2}^1 5t^4 dt = (1)^5 + 32 = 33$$

$$A(x) = \int_{-2}^x 5t^4 dt = x^5 + 32$$



$$A(2) = \int_{-2}^2 5t^4 dt = (2)^5 + 32 = 64$$

$$A(x) = \int_{-2}^x 5t^4 dt = x^5 + 32$$



$$A(3) = \int_{-2}^3 5t^4 dt = (3)^5 + 32 = 275$$

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- ▶ Guess a function with derivative $f(x)$: $F(x)$.
- ▶ Then $A(x) = F(x) + C$ for some constant C .
- ▶ Also $A(a) = 0$, so $0 = F(a) + C$, so $C = -F(a)$
- ▶ So, $A(x) = F(x) - F(a)$

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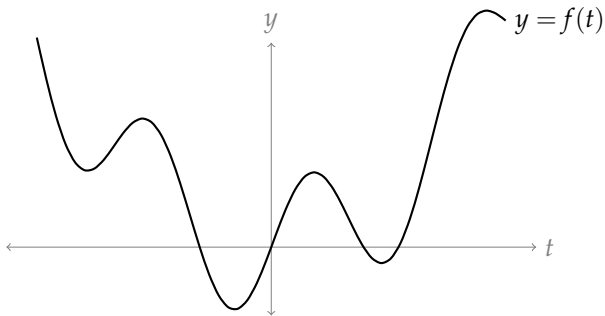
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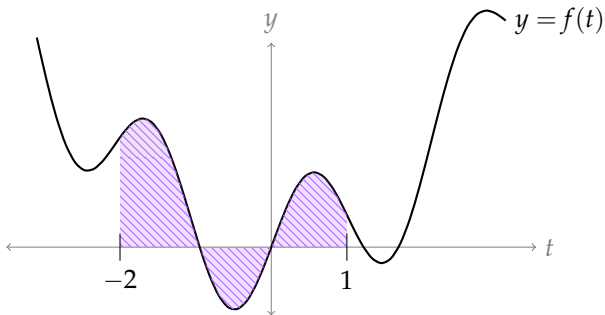
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$$\int_a^b f(t) \, dt = F(b) - F(a) \quad \text{where} \quad F'(x) = f(x)$$

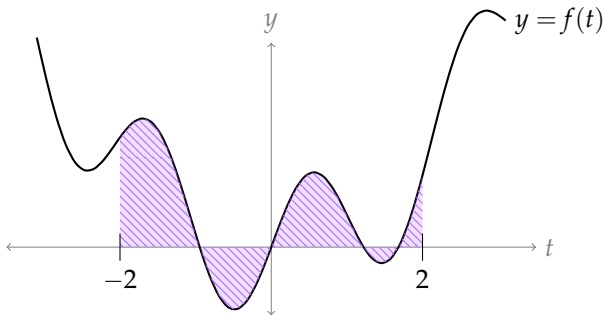


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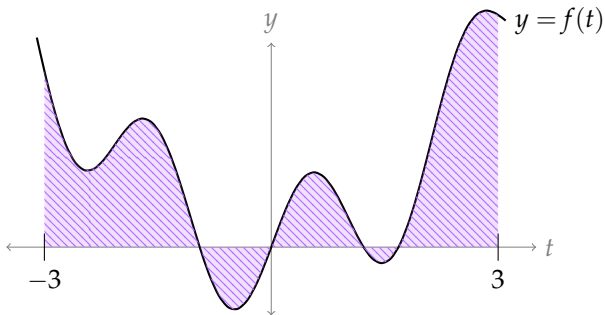
$$\int_{-2}^1 f(t) \, dt = F(1) - F(-2)$$

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$$\int_{-3}^3 f(t) \, dt = F(3) - F(-3)$$

Fundamental Theorem of Calculus, Part 2

Let $F(x)$ be differentiable, defined, and continuous on the interval $[a, b]$ with $F'(x) = f(x)$ for all $a < x < b$. Then

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Examples:

$$\frac{d}{dx} \{5x^7\} = 35x^6, \text{ so}$$

$$\int_0^3 35x^6 \, dx =$$

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$$\int_0^3 35x^6 \, dx = 5x^7 \Big|_{x=3} - 5x^7 \Big|_{x=0} = 5(3^7) - 5(0^7) = 5 \cdot 3^7$$

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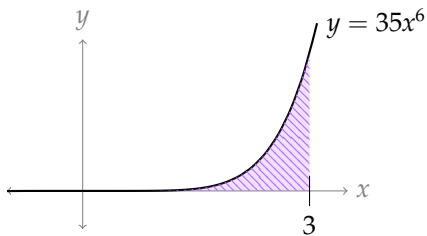
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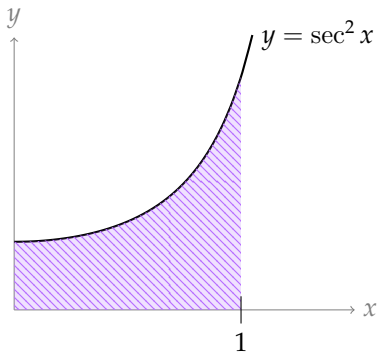
$$\int_0^{\pi/4} \sec^2 x \, dx = \tan x \Big|_{x=\pi/4} - \tan x \Big|_{x=0} = \tan(\pi/4) - \tan 0 = 1$$

$$\int_0^3 35x^6 \, dx = F(b) - F(a) \quad \text{where} \quad F(x) = 5x^7$$



$$\int_0^3 35x^6 \, dx = 5(3)^7 - 5(0)^7$$

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RELEVANT VOCABULARY

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If $F(x)$ is a function whose derivative is $f(x)$, we call $F(x)$ an **antiderivative** of $f(x)$.

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An antiderivative of $\sin x$ is $-\cos x$, because $\frac{d}{dx} \{-\cos x\} = \sin x$.



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The function $f(x)$ evaluated from a to b

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The function $f(x)$ evaluated from a to b

FTC Part 2, Abridged

$$\int_a^b f(x) \, dx = F(x) \Big|_a^b$$

where $F(x)$ is an antiderivative of $f(x)$

Definition

The **indefinite integral** of a function $f(x)$:

$$\int f(x) \, dx$$

means the *most general* antiderivative of $f(x)$.

Examples:

$$\int 2x \, dx =$$

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$$\int 2x \, dx = x^2 + C,$$

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The **indefinite integral** of a function $f(x)$:

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Remember: two functions with the same derivative differ by a constant, so we include the “+C” for indefinite integrals.

DEFINITE VS INDEFINITE INTEGRALS

For each pair of properties below, decide which applies to **definite** integrals, and which to **indefinite** integrals.

No limits (or bounds) of integration, $\int f(x) dx$	
Limits (or bounds) of integration, $\int_a^b f(x) dx$	
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ANTIDIFFERENTIATION BY INSPECTION

1. $\int e^x \, dx$

ANTIDIFFERENTIATION BY INSPECTION

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$$7. \int nx^{n-1} dx \quad (n \neq 0, \text{ constant})$$



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$$8. \int x^n dx \quad (n \neq -1, \text{ constant})$$



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$$7. \int nx^{n-1} dx = x^n + C \quad (n \neq 0, \text{ constant})$$

$$8. \int x^n dx = \frac{1}{n+1}x^{n+1} + C \quad (n \neq -1, \text{ constant})$$



Power Rule for Antidifferentiation

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C$$

if $n \neq -1$ is a constant

Example:

$$\int (5x^2 - 15x + 3) dx =$$



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Example:

$$\int (5x^2 - 15x + 3) dx = \frac{5}{3}x^3 - \frac{15}{2}x^2 + 3x + C$$



ANTIDERIVATIVES TO RECOGNIZE

- ▶ $\int x^n dx = \frac{1}{n+1}x^{n+1} + C$
- ▶ $\int a dx = ax + C$
- ▶ $\int e^x dx = e^x + C$
- ▶ $\int \frac{1}{x} dx = \log |x| + C$
- ▶ $\int \sin x dx = -\cos x + C$
- ▶ $\int \cos x dx = \sin x + C$
- ▶ $\int \sec^2 x dx = \tan x + C$
- ▶ $\int \sec x \tan x dx = \sec x + C$
- ▶ $\int \csc x \cot x dx = -\csc x + C$
- ▶ $\int \csc^2 x dx = -\cot x + C$
- ▶ $\int \frac{1}{1+x^2} dx = \arctan x + C$
- ▶ $\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$

Included Work



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